break down— no matter how esoteric the realm might seem—throws up red flags. If the known laws of physics break down under any circumstances, it is a clear signal that we have not reached the deepest possible understanding. After all, the universe works; as far as we can tell, the universe does not break down. The correct theory of the universe should, at the very least, meet the same standard.

Well, that surely seems reasonable. But for my money, the full urgency of the conflict between general relativity and quantum mechanics is revealed only through another example. Look back at Figure 10.6. As you can see, we have made great strides in piecing together a consistent and predictive story of cosmic evolution, but the picture remains incomplete because of the fuzzy patch near the inception of the universe. And within the foggy haze of those earliest moments lies insight into the most tantalizing of mysteries: the origin and fundamental nature of space and time. So what has prevented us from penetrating the haze? The blame rests squarely on the conflict between general relativity and quantum mechanics. The antagonism between the laws of the large and those of the small is the reason the fuzzy patch remains obscure and we still have no insight into what happened at the very beginning of the universe.

To understand why, imagine, as in Chapter 10, running a film of the expanding cosmos in reverse, heading back toward the big bang. In reverse, everything that is now rushing apart comes together, and so as we run the film farther back, the universe gets ever smaller, hotter, and denser. As we close in on time zero itself, the *entire* observable universe is compressed to the size of the sun, then further squeezed to the size of the earth, then crushed to the size of a bowling ball, a pea, a grain of sand— smaller and smaller the universe shrinks as the film rewinds toward its initial frames. There comes a moment in this reverse-run film when the entire known universe has a size close to the Planck length— the millionth of a billionth of a billionth of a billionth of a centimeter at which general relativity and quantum mechanics find themselves at loggerheads. At this moment, all the mass and energy responsible for spawning the observable universe is contained in a speck that’s less than a hundredth of a billionth of a billionth of the size of a single atom.[8]